Student number:

Semester 2, 2024

Mathematics and Statistics

MAST20030_24_SM2 - Differential Equations

Reading time: 15 minutes

Writing time: 3 hours

Exam Venue Rules

- Do not communicate with other students in the exam venue
- During writing time, write your student number on the cover sheet, and on any script books or answer pages
- Blue scrap paper is for informal working notes only and will be collected and destroyed at the end of the session
- No part of the exam paper may be removed from the venue
- Ensure you only have permitted materials on your desk. The permitted materials are listed on this cover sheet
- Raise your hand if you require assistance
- Turn off your phone prior to commencement of reading time. Phones must be fully powered off

Standard Permitted Materials

- Identification card (Hardcopy student ID, Passport or Australian Government-issued photo ID)
- Pencils, pens, erasers
- Clear or unlabelled bottle of water

Additional Permitted Materials

- No other permitted materials

Exam Instructions

- During reading time, students may not write, type, or select answers
- Students must use a blue or black ball-point pen
- Students may use a pencil for diagrams
- Show all working or calculation steps for each question
- Write your answers in the designated answer boxes provided for each question
- If you require additional space for your answers, use the pages at the back of the exam paper
- Marks and weightings differ per question. Ensure you verify the marks for each question
- Next to each question you will find its assigned marks/weight.

Special Exam Instructions

None

Exam Weighting

None

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Question 1 (24 marks)

Consider the system

$$\begin{bmatrix} x'(t) \\ y'(t) \end{bmatrix} = \boldsymbol{A} \cdot \begin{bmatrix} x(t) \\ y(t) \end{bmatrix} \qquad \text{where} \qquad \boldsymbol{A} = \begin{bmatrix} -\alpha & 1 \\ 1 & -\alpha \end{bmatrix}, \quad \alpha \in \mathbb{R}.$$

(a) Find the eigenvalues of $\boldsymbol{A}$.

(b) Identify all the possible type and stability of the critical point $(0, 0)$ for all non-zero eigenvalues of $\boldsymbol{A}$.

(c) Consider the system when $\alpha = 0$.

(i) Express the system as a single 2nd order differential equation for $x(t)$.

(ii) Given the system has the solution

$$\begin{bmatrix} x \\ y \end{bmatrix} = C_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} e^t + C_2 \begin{bmatrix} -1 \\ 1 \end{bmatrix} e^{-t},$$

sketch its phase portrait.

Include the following information and all relevant calculations and justification:

- any straight line orbits and their directions;
- at least 4 other orbits and their directions, showing the asymptotic behaviour as $t \to \infty$ and $t \to -\infty$;
- the slopes at which the orbits cross the $x-$ and $y-$axes;
- the loci of all minima and maxima in x, and also in y, if any of these exist;
- the orbit (or orbits) where the ratio y/x remains constant for all t and the value of this ratio;
- the value of $y(0)$, given $x(0) = a > 0$ and $\lim_{t\to\infty}(x, y) = (0, 0)$;
- the type and stability of the critical point at $(0, 0)$.

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Question 2 (8 marks)

Consider the IVP

$$(t^2 + At + B)\ y''(t) + 2t\ y'(t) - 2y(t) = g(t),$$
$$y(1) = y'(1) = \pi.$$

(a) Let $A = 8, \quad B = 7, \quad g = \frac{e^t}{(2-t)}$.

Find the largest interval D where the IVP is guaranteed a unique solution that is twice differentiable.

(b) Let $A = B = 0, \quad g = 0$.

Given $v(t) = t$ is a solution, use reduction of order to solve the IVP.

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Question 3 (16 marks)

Use Laplace transforms to find $y(t)$ for each of the following.

(a)

$$y''(t) + 9y(t) = f(t)$$

(i) $f(t)$ is a unit impulse force applied at $t = 1$ and $y(0) = 0, \quad y(\pi/4) = 0$.

(ii) $f(t)$ is a unit force that is switched on at $t = 1$ and $y(0) = 0, \quad y(\pi/2) = \sin(3)$.

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(b)

$$y''(t) \;+\; \int_0^t y'(\tau)e^{2(t-\tau)}d\tau \;=\; e^{2t}$$

$$y(0) = y'(0) = 0.$$

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Question 4 (12 marks)

Use power series to solve for $y(x)$ around $x = 0$, given

$$(x^2 + 1)y'' - 4xy' + 6y = 0.$$

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Question 5 (30 marks)

A rod of unit length is initially heated to a temperature of $f(x)$. The rod is insulated except at its ends. The temperature distribution along the rod $\phi(x,t)$, for a given position $x \in [0,1]$ and time $t \in (0,\infty)$, is governed by

$$\frac{\partial \phi}{\partial t} = \frac{\partial^2 \phi}{\partial x^2}$$

subject to

$$\phi(0,t) = 0, \qquad \frac{\partial \phi}{\partial x}(1,t) = 1, \qquad t > 0$$

$$\phi(x,0) = f(x) = x + 5\sin\left(\frac{5\pi x}{2}\right), \quad 0 \le x \le 1.$$

Assume

$$\phi(x,t) = \phi_s(x) + v(x,t).$$

(a) Find the steady state temperature ϕ_s.

(b) Write down the boundary conditions for the transient temperature v.

(c) Use the method of Separation of Variables to find the general solution for v.

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Continued answer space for Question 5(c)

(d) Find the solution for ϕ.

(e) Sketch $\phi(x, 0)$, $\phi(x, t_1)$, $\phi(x, t_2)$ and $\phi(x, \infty)$, $0 < t_1 < t_2 < \infty$ on one set of axes.

(f) Consider $\lim_{t\to\infty} \phi(x,t)$.

(i) Find the temperature at the midpoint of the rod.

(ii) Find the total heat in the rod. Not in 2025

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Question 6 (18 marks)

(a) Consider

$$f(x) = \begin{cases} 0 & -\pi < x < -\frac{\pi}{2} \\ \sin x & -\frac{\pi}{2} \leq x \leq \frac{\pi}{2} \\ 0 & \frac{\pi}{2} < x \leq \pi. \end{cases}$$

Find the Fourier series of $f(x),\ x \in (-\pi, \pi]$.

(b) The linear operator

$$\mathsf{T} = -\frac{d^2}{dx^2} + 2\frac{d}{dx}$$

with the boundary conditions

$$y'(0) = y'(1) = 0$$

has eigenfunctions

$$y_n(x) = \mathrm{e}^x \cos n\pi x, \quad n = 1, 2, 3, ...$$

Find the eigenvalue λ_n for each eigenfunction y_n.

(c) The Fourier series of $x^4, x \in (-\pi, \pi]$ is given by

$$s(x) = \frac{\pi^4}{5} + \sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \left(8\pi^2 - \frac{48}{n^2}\right) \cos nx.$$

Use this result to find

$$\sum_{n=1}^{\infty} \left(\frac{1}{n^4} - \frac{\pi^2}{6}\frac{1}{n^2}\right).$$

(d) Consider

$$y''(x) - 2y(x) = s(x)$$

subject to periodic boundary conditions at $x = \pm\pi$.

Given $s(x)$ is the Fourier series of $x^2,\ x \in (-\pi, \pi]$,

$$s(x) = \frac{\pi^2}{3} + 4\sum_{n=1}^{\infty} \frac{(-1)^n}{n^2} \cos nx,$$

find a *particular solution* to the ODE by trialling an arbitrary Fourier series. **You are not required to find the complementary solution.**

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Question 7 (12 marks)

(a) Use the integral definition of the Fourier transform to evaluate

$$\mathscr{F}\Big(\delta(x-a)\Big), \qquad a \in \mathbb{R}.$$

(b) Consider the 1D advection PDE for $u(x,t)$,

$$\frac{\partial u}{\partial t} = c\frac{\partial u}{\partial x}, \qquad x \in \mathbb{R},\ t > 0,\ c \in \mathbb{R},$$

subject to the initial condition $u(x,0) = \delta(x)$.

Use Fourier transforms to find u.

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(c) Use the following Fourier transform

$$\mathscr{F}\left(\mathrm{e}^{-ax^2}\right) = \sqrt{\frac{\pi}{a}}\ \mathrm{e}^{-\frac{k^2}{4a}}, \qquad a > 0,$$

to evaluate

$$\int_{-\infty}^{\infty} \cos(2x)\ \mathrm{e}^{-x^2/2} dx.$$

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End of Exam — Total Available Marks = 120

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MAST20030 Differential Equations Formulae Sheet

1) **Laplace Transforms**

1.	$f(t)$	$F(s) = \mathscr{L}(f) = \int_0^\infty f(t)e^{-st}\,dt$	(Definition of Transform)
2.	1	$\frac{1}{s}$	
3.	t^n	$\frac{n!}{s^{n+1}}$	
4.	e^{at}	$\frac{1}{s-a}$	
5.	$\sin(at)$	$\frac{a}{s^2+a^2}$	
6.	$\cos(at)$	$\frac{s}{s^2+a^2}$	
7.	$\sinh(at)$	$\frac{a}{s^2-a^2}$	
8.	$\cosh(at)$	$\frac{s}{s^2-a^2}$	
9.	$\delta(t-a)$	e^{-as}	$(a \geq 0)$
10.	$f'(t)$	$sF(s) - f(0)$	
11.	$f''(t)$	$s^2F(s) - sf(0) - f'(0)$	
12.	$f^{(n)}(t)$	$s^nF(s) - \sum_{k=0}^{n-1} s^{n-1-k} f^{(k)}(0)$	
13.	$\int_0^t f(\tau)\,d\tau$	$\frac{F(s)}{s}$	
14.	$e^{-at}f(t)$	$F(s+a)$	(s-Shifting Theorem)
15.	$f(t-a)\,u(t-a)$	$e^{-as}F(s)$	($a > 0$, t-Shifting Theorem) $u(t)$ is the Heaviside step function
16.	$\int_0^t f(\tau)\,g(t-\tau)\,d\tau$	$F(s)\,G(s)$	(Convolution)

2) **Standard Limits**

(i) $\lim_{n\to\infty} \frac{1}{n^p} = 0 \quad (p > 0)$

(ii) $\lim_{n\to\infty} r^n = 0 \quad (|r| < 1)$

(iii) $\lim_{n\to\infty} a^{1/n} = 1 \quad (a > 0)$

(iv) $\lim_{n\to\infty} n^{1/n} = 1$

(v) $\lim_{n\to\infty} \frac{a^n}{n!} = 0 \quad (\text{all } a)$

(vi) $\lim_{n\to\infty} \frac{\log_e n}{n^p} = 0 \quad (p > 0)$

(vii) $\lim_{n\to\infty} \left(1 + \frac{a}{n}\right)^n = \mathrm{e}^a \quad (\text{all } a)$

(viii) $\lim_{n\to\infty} \frac{n^p}{a^n} = 0 \quad (\text{all } p, a > 1)$

3) **The Generalised Harmonic Series (p-Series)**

$$\sum_{n=1}^{\infty} \frac{1}{n^p} \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } p > 1 \\ \text{divergent} & \text{if } p \leq 1 \end{cases}$$

4) **Geometric Series**

$$\sum_{n=0}^{\infty} ar^n \quad \text{is} \quad \begin{cases} \text{convergent} & \text{if } |r| < 1 \\ \text{divergent} & \text{if } |r| \geq 1 \end{cases}$$

5) **Fourier Series Formulae**

$$\begin{aligned} f(x) &= \frac{1}{2}a_0 + \sum_{n=1}^{\infty} \left(a_n \cos k_n x + b_n \sin k_n x\right), \quad k_n = \frac{n\pi}{L} \\ a_0 &= \frac{1}{L}\int_{-L}^{L} f(x)\, dx \\ a_n &= \frac{1}{L}\int_{-L}^{L} f(x) \cos k_n x\, dx \\ b_n &= \frac{1}{L}\int_{-L}^{L} f(x) \sin k_n x\, dx \end{aligned}$$

6) **Fourier Transforms Formulae**

$$\widehat{f}(k) = \mathscr{F}(f) = \int_{-\infty}^{\infty} f(x)\mathrm{e}^{-ikx}\, dx$$

$$f(x) = \mathscr{F}^{-1}(\widehat{f}) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \widehat{f}(k)\mathrm{e}^{ikx}\, dk$$

$$\mathscr{F}\left(\frac{\mathrm{d}^n f}{\mathrm{d}x^n}\right) = (ik)^n \widehat{f}, \qquad \mathscr{F}\left(x^n f\right) = i^n \frac{\mathrm{d}^n \widehat{f}}{\mathrm{d}k^n}$$

7) **Trigonometric and Hyperbolic Formulae**

$$\cos^2 x + \sin^2 x = 1$$
$$1 + \tan^2 x = \sec^2 x$$
$$\cot^2 x + 1 = \operatorname{cosec}^2 x$$

$$\cosh^2 x - \sinh^2 x = 1$$
$$1 - \tanh^2 x = \operatorname{sech}^2 x$$
$$\coth^2 x - 1 = \operatorname{cosech}^2 x$$

$$\sin(x+y) = \sin x \cos y + \cos x \sin y$$
$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\sinh(x+y) = \sinh x \cosh y + \cosh x \sinh y$$
$$\cosh(x+y) = \cosh x \cosh y + \sinh x \sinh y$$

$$\sin x \sin y = \frac{1}{2}\left[\cos(x-y) - \cos(x+y)\right]$$
$$\cos x \cos y = \frac{1}{2}\left[\cos(x-y) + \cos(x+y)\right]$$
$$\sin x \cos y = \frac{1}{2}\left[\sin(x-y) + \sin(x+y)\right]$$

$$\sinh x \sinh y = \frac{1}{2}\left[\cosh(x+y) - \cosh(x-y)\right]$$
$$\cosh x \cosh y = \frac{1}{2}\left[\cosh(x+y) + \cosh(x-y)\right]$$
$$\sinh x \cosh y = \frac{1}{2}\left[\sinh(x+y) + \sinh(x-y)\right]$$

$$\sin x = \frac{i}{2}\left(\mathrm{e}^{-ix} - \mathrm{e}^{ix}\right)$$
$$\cos x = \frac{1}{2}\left(\mathrm{e}^{-ix} + \mathrm{e}^{ix}\right)$$

$$\sinh x = \frac{1}{2}\left(\mathrm{e}^{x} - \mathrm{e}^{-x}\right)$$
$$\cosh x = \frac{1}{2}\left(\mathrm{e}^{x} + \mathrm{e}^{-x}\right)$$

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